ORBION AGENTS

Mobile Application Architecture & Interactive Prototype Specification

Complete End-to-End Screen Inventory, Interaction Design, Data Contracts, WhatsApp Templates & Backend Traceability

Document Purpose: Executive Architecture Review & Mobile Prototype Demo

Target Platforms: iOS (App Store / TestFlight) & Android (Google Play)

Verified Backend: FastAPI Python 3.11, PostgreSQL, Redis, Celery

Interactive Prototype File: mobile-prototype/index.html (17 Screens + Modal)

Date: October 2026 · Release v1.0

Recommended Stack: React Native + Expo SDK 51 (JavaScript)

State Management: TanStack Query v5 + Zustand

Secure Storage: Hardware Keychain/Keystore (expo-secure-store)

Scope Notice: Brain / AI Workspace excluded from MVP

Author: Senior Mobile Architect & Product UX Team

1. Executive Summary & Mobile Architecture

This specification document details the end-to-end mobile application architecture proposal and interactive prototype for **Orbion Agents**, designed for management review. The mobile application reuses the existing production backend (FastAPI, Redis pub/sub, PostgreSQL, Celery workers) without duplicating backend business logic or database schemas.

Primary Business Objectives:
• **Real-Time Customer Responsiveness:** Enable business owners, sales leads, and support agents to triage and reply to incoming WhatsApp, Instagram, and Twilio customer inquiries on-the-go.
• **Operational Continuity:** Monitor automated AI agents, toggle automation flows, view live pipeline metrics, and manage WhatsApp message templates from a thumb-optimized mobile interface.
• **Cost Efficiency:** Leverage React Native with Expo to maximize shared JavaScript conventions with the existing Next.js frontend, enabling rapid development and zero app store submission friction for JavaScript OTA updates.

Critical Scope Boundary (Mandatory Exclusion):
As directed, **Brain, RAG knowledge bases, document ingestion, URL/sitemap crawlers, and the AI Workspace canvas are 100% excluded** from all mobile navigation bars, bottom tabs, screen inventories, and prototype interactions in this phase.

ARCHITECTURE SUMMARY	VERIFIED REPOSITORY FINDINGS
Mobile Framework: React Native with Expo SDK 51+ **Language:** JavaScript ES2023 (strictly matches repo jsconfig.json) **Navigation:** React Navigation 6 (5 Bottom Tabs + Stack + Modals) **Bottom Tabs:** 1. Home | 2. Inbox | 3. Dashboard | 4. Agents | 5. More **Client State:** Zustand (lightweight global auth & workspace context) **Server State:** TanStack Query v5 (offline caching & optimistic mutations) **Token Storage:** expo-secure-store (Keychain iOS / Keystore Android) **Realtime:** Native WebSocket to /ws/{user_id} with Redis Pub/Sub **Theme:** Premium Dark SaaS (#05080e, Accent #814AC8)	**Auth Endpoints:** POST /auth/send-otp, /auth/verify-otp, Google OAuth **Conversations API:** GET /conversations, /messages/{id}, POST /send-reply **AI Suggestions:** POST /ai-suggest live in composer **Templates API:** GET /templates, POST /templates/submit/{id}, /messages/send **Automation Flows:** GET /flows, PATCH /flows/{id}/status (Active/Paused) **CRM Leads:** GET /lead-scoring/leads with automated AI scoring **Gaps Identified:** No FCM/APNs push service (gap); 30-min JWT expiry has no refresh endpoint; Followups router commented out in backend.

SCR-000 — Splash Screen & Session Restoration

Module: App Initialization | Priority: P0 (MVP)



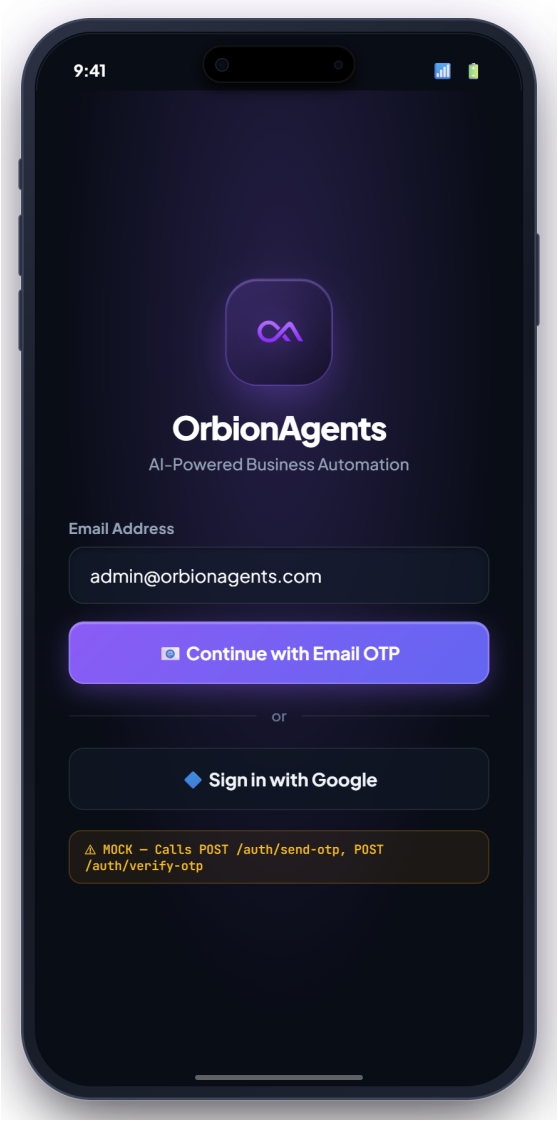
Purpose:	First visual touchpoint during cold starts. Restores secure session tokens from hardware SecureStore, performs token validation with FastAPI, establishes WebSocket connectivity, and smoothly transitions to Dashboard or Login.
User Entry:	App launch from home screen or cold start.
Primary UI:	Ambient glowing background rings, 3D animated Orbion brand icon with spinning conic gradient border, App brand title & pill, Status ticker bar with animated shimmering progress, 'Get Started' action button, Security & Encryption badges.
User Actions:	Cold start → Read expo-secure-store auth_token → Validate /auth/me → If valid, connect /ws and navigate to SCR-004 Dashboard; if expired/empty, navigate to SCR-001 Login.

API Endpoints:	`expo-secure-store.getItemAsync('auth_token')`
	`GET /auth/me`
	`GET /onboarding/status`

States:	Loading: Animated progress bar + dynamic status ticker ('Restoring token...', 'Connecting WebSocket...'). Error/Expired: Seamlessly transitions to SCR-001 Login.
RBAC Permission:	Public / System Cold Start.
Mobile UX Notes:	Sleek animated splash screen prevents white flash on boot. Native splash hides seamlessly when first React render tree completes.

SCR-001 — Login (Email OTP & Google Auth)

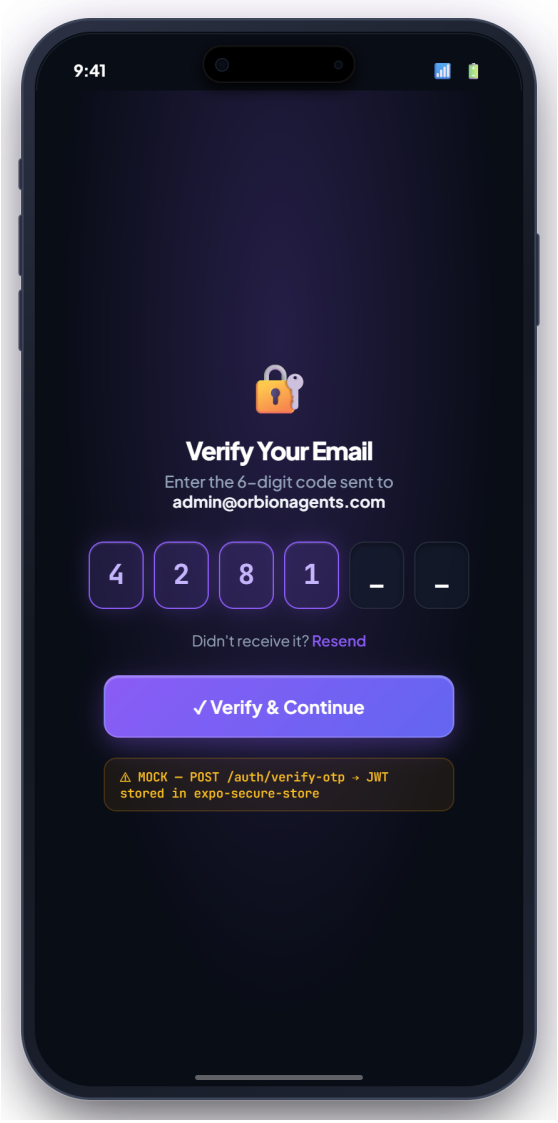
Module: Authentication | Priority: P0 (MVP)



Purpose:	First-time authentication gateway. Allows users to request a one-time passcode (OTP) via email or initiate single sign-on via Google OAuth.
User Entry:	App launch / unauthenticated session / session expiration.
Primary UI:	App branding logo, Email address text field, 'Continue with Email OTP' primary CTA button, Divider, 'Sign in with Google' secondary button.
User Actions:	Enter email → POST /auth/send-otp → Navigate to SCR-002 OTP Verification. Tap Google → Launch browser OAuth session.
API Endpoints:	`POST /auth/send-otp` `GET /auth/google/login`
States:	Loading: Spinner on button during email dispatch. Error: Inline toast for invalid domain or rate limiting. Success: Transition to OTP screen.
RBAC Permission:	Public / Unauthenticated.
Mobile UX Notes:	Keyboard-aware view with auto-capitalization disabled for email. Hardware-level autofill supported.

SCR-002 — 6-Digit Email OTP Verification

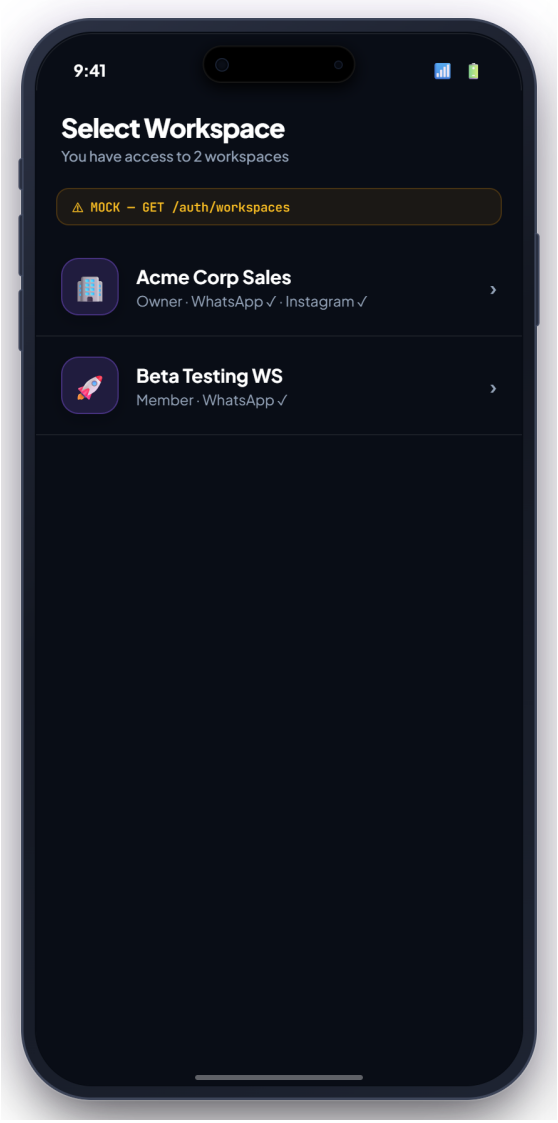
Module: Authentication | Priority: P0 (MVP)



Purpose:	Validates the one-time passcode sent to the user's email, exchanges the code for a signed JWT access token, and securely stores credentials.
User Entry:	After successful email submission from SCR-001.
Primary UI:	6 individual OTP numeric entry boxes, targeted email label, Resend code countdown timer, 'Verify & Continue' primary action button.
User Actions:	Input 6 digits → POST /auth/verify-otp → Receive JWT token → Save to expo-secure-store → Check 2FA requirement → Navigate to Workspace or Dashboard.
API Endpoints:	`POST /auth/verify-otp` `POST /2fa/verify-login (if 2FA enabled)`
States:	Loading: Auto-submits on 6th digit. Error: Red border on boxes with 'Invalid or expired code'. Success: Haptic success pulse.
RBAC Permission:	Public / Pre-authenticated session.
Mobile UX Notes:	SMS/Email OTP auto-read supported via iOS TextInput textContentType='oneTimeCode'. Auto-focuses next digit box on input.

SCR-003 — Multi-Tenant Workspace Selection

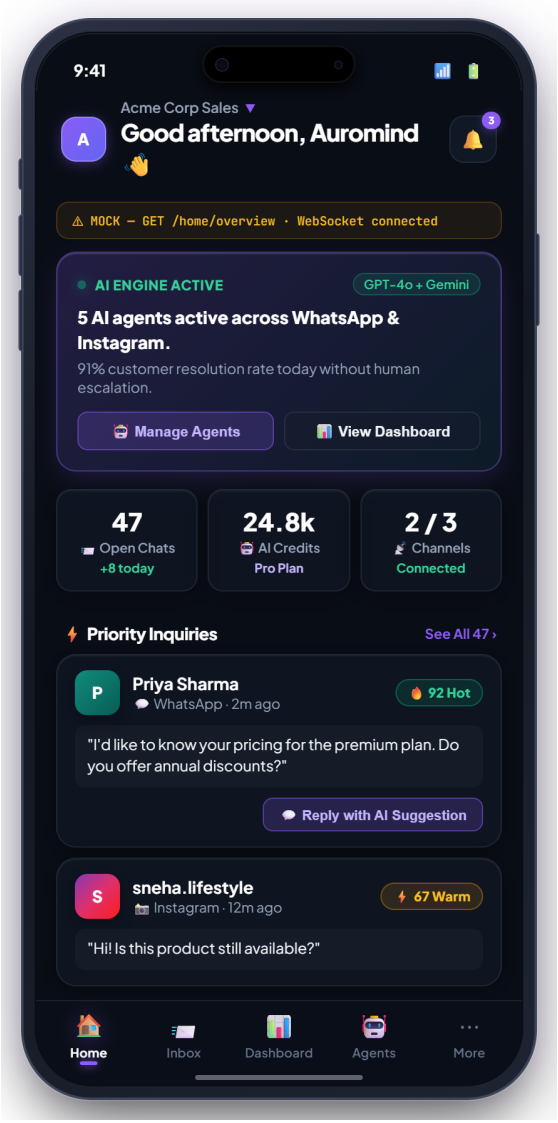
Module: Workspace Context | Priority: P0 (MVP)



Purpose:	Displays all workspaces the authenticated user belongs to. Enforces strict tenant isolation by activating the selected workspace context for all API and WebSocket operations.
User Entry:	Post-login for multi-workspace users, or via 'Switch Workspace' in More menu.
Primary UI:	Header with available count, Workspace cards displaying workspace name, icon, user's assigned role (Owner/Admin/Member), and active channel indicators.
User Actions:	Tap workspace card → Save workspace_id in Zustand + storage → Initialize WebSocket room subscription → Route to Home Hub.
API Endpoints:	`GET /auth/workspaces` `GET /workspaces/{id}/my-permissions`
States:	Loading: Skeleton cards. Error: Alert with retry option. Single workspace: Automatically bypassed directly into Home Hub.
RBAC Permission:	Authenticated user.
Mobile UX Notes:	Crucial for agency operators managing multiple business client workspaces from a single phone.

SCR-003B — Executive Mobile Home Hub

Module: Home & Overview | Priority: P0 (MVP)



Purpose:	Primary executive home hub. Delivers an instant pulse on autonomous AI operations, live omnichannel agent statuses, unread customer inquiries, and quick-launch shortcuts.
User Entry:	Bottom Tab #1 (Home icon) or post-login/workspace selection default landing.
Primary UI:	Header with avatar, workspace context chip, greeting & notification bell; Live AI Copilot status card; 3-column Quick Pulse stats (Open Chats, AI Credits, Channel Count); Priority Inquiries (Hot Leads) action cards; Quick Launch app grid; Channel Health indicators.
User Actions:	Tap 'Reply with AI Suggestion' → Open conversation (SCR-006); Tap 'Manage Agents' → Open AI Agents (SCR-007); Tap Quick Launch tiles → Open respective modules; Tap bottom tabs to navigate.

API Endpoints:	`GET /home/overview`
	`GET /conversations/counts`
	`GET /channels/status`
	`WebSocket: live_agent_events`

States:	Online: Green pulsing indicator with active model tags (GPT-4o + Gemini). Offline/Reconnecting: Amber banner with retry.
RBAC Permission:	Authenticated user with active workspace.
Mobile UX Notes:	Serves as the central command center for busy executives and business owners, reducing navigation depth to 1-tap for 80% of daily workflows.

SCR-004 — Executive Business Metrics Dashboard

Module: Dashboard & KPIs | Priority: P0 (MVP)



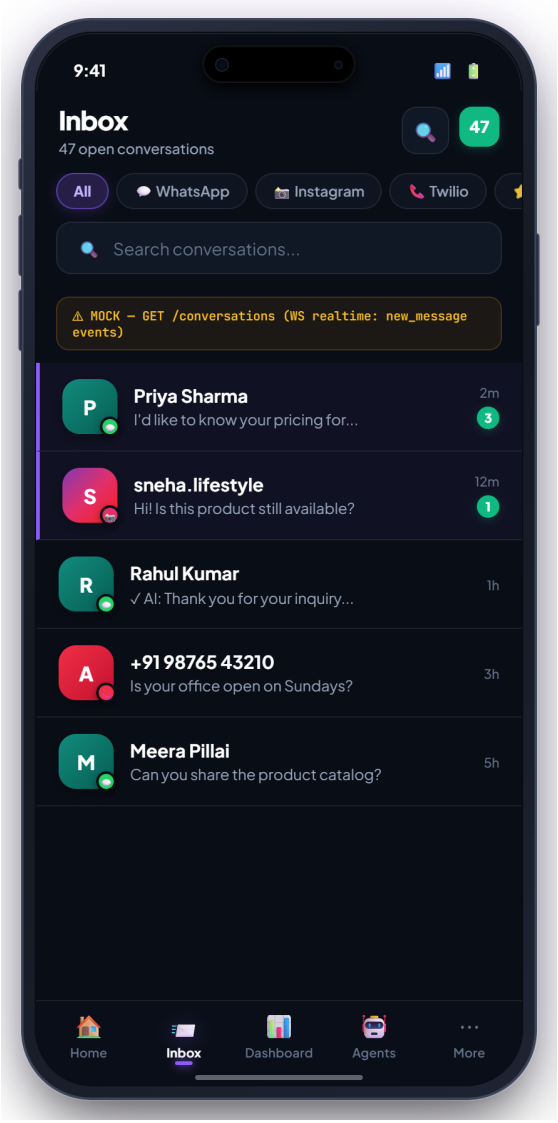
Purpose:	High-level overview of daily business operations. Surfaces unread customer volume, AI resolution efficacy, active campaigns, wallet balances, and rapid quick actions.
User Entry:	Bottom Tab #3 (Dashboard icon) or Quick Launch from Home Hub.
Primary UI:	Top greeting with Notification Bell & Avatar; Credits widget (AI credits & WCC ■ balance); 2x2 Metric KPI cards; 6-button Quick Actions grid; Recent Activity feed.
User Actions:	Tap Quick Action (Inbox, Templates, Campaigns, Contacts, Agents, Flows) to navigate; Tap notification bell to open SCR-013; Pull down to refresh all metrics.

API Endpoints:	`GET /dashboard/overview`
	`GET /dashboard/metrics`
	`GET /dashboard/activities`
	`GET /dashboard/insights`

States:	Loading: Staggered pulse shimmer. Empty: Zero-state guides to connect channels. Success: Animated counter increments.
RBAC Permission:	Requires permission: dashboard.overview.
Mobile UX Notes:	Replaces heavy desktop Three.js graphs with high-contrast, glanceable native summary cards optimized for outdoor readability.

SCR-005 — Unified Omnichannel Inbox

Module: Unified Inbox | Priority: P0 (MVP)



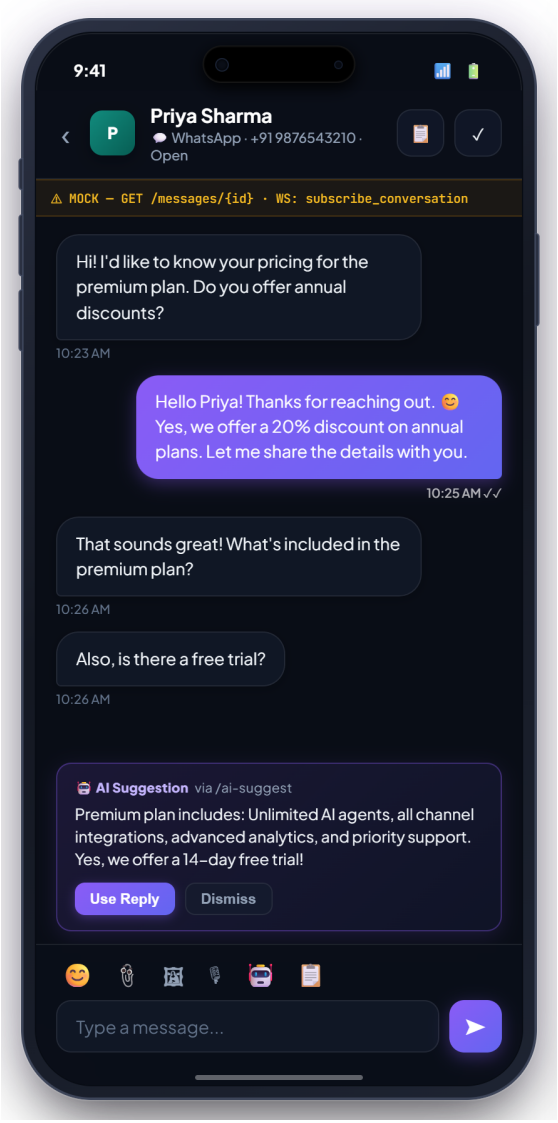
Purpose:	Central hub for customer communications across WhatsApp Business, Instagram DM, and Twilio. Displays conversation list with unread counts and channel identity badges.
User Entry:	Bottom Tab #1 (Inbox tray icon) or Quick Action from Dashboard.
Primary UI:	Header with open conversation counter; Channel filter horizontal pill chips (All, WhatsApp, Instagram, Twilio, Starred); Search bar; Conversation item rows with channel avatars.
User Actions:	Tap filter chip to filter channels; Tap conversation row to open SCR-006; Swipe row to resolve/close; Pull-to-refresh list.

API Endpoints:	`GET /conversations?workspace_id={id}&channel;={ch}`
	`GET /conversations/counts`
	`WebSocket: new_message event`

States:	Loading: Shimmer rows. Empty: 'No conversations in this filter'. Realtime: New message increments counter with green ping.
RBAC Permission:	Requires permission: inbox.conversations.
Mobile UX Notes:	High-performance FlatList with windowSize=10 and getItemLayout to ensure 60fps scrolling across 1,000+ active conversations.

SCR-006 — Live Customer Chat & AI Suggestions

Module: Unified Inbox | Priority: P0 (MVP)



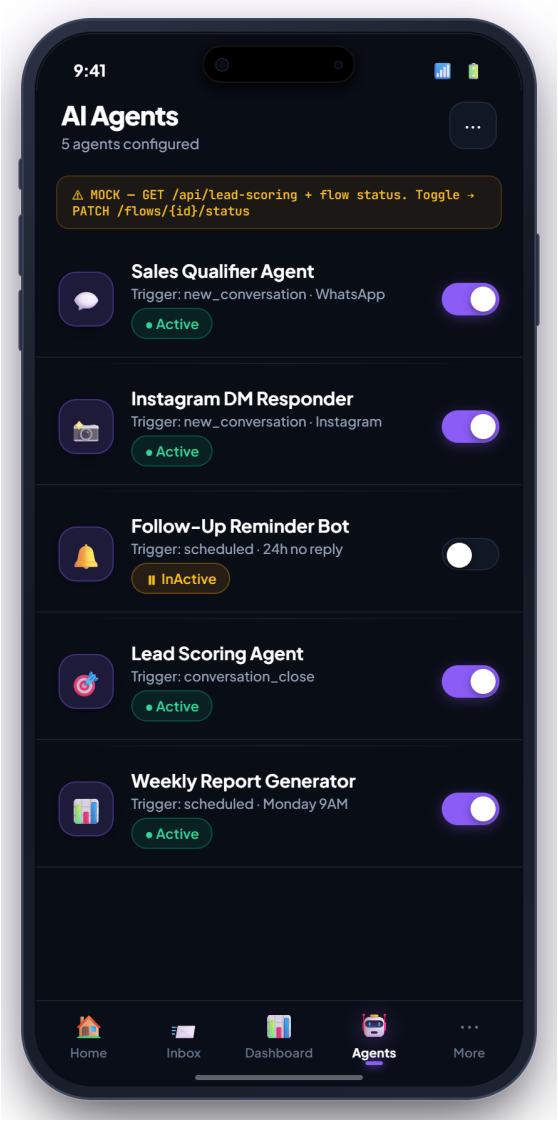
Purpose:	Full-duplex conversation thread for a customer inquiry. Displays message history, channel identity, AI-suggested smart replies, and omnichannel message composer.
User Entry:	Tap any conversation row in SCR-005 Inbox.
Primary UI:	Header with customer name, phone number, channel badge, resolve button; Scrollable message history bubble thread; AI Suggestion card; Reply composer with toolbar (Emoji, Attach, Templates, AI).
User Actions:	Type reply → POST /send-reply; Tap 'Use Reply' on AI suggestion → Auto-fill composer; Tap '■' icon → Open Message Templates (SCR-009); Tap '✓' → Mark conversation resolved.

API Endpoints:	`GET /messages/{conversation_id}`
	`POST /send-reply`
	`POST /ai-suggest`
	`POST /conversations/{id}/close`

States:	Optimistic send: Message appears immediately with clock icon, updates to double checkmark upon WebSocket server confirmation.
RBAC Permission:	Requires permission: inbox.conversations.
Mobile UX Notes:	KeyboardAvoidingView with behavior='padding' on iOS and 'height' on Android. Auto-scrolls to newest message on focus.

SCR-007 — AI Agents Status & Runtime Toggles

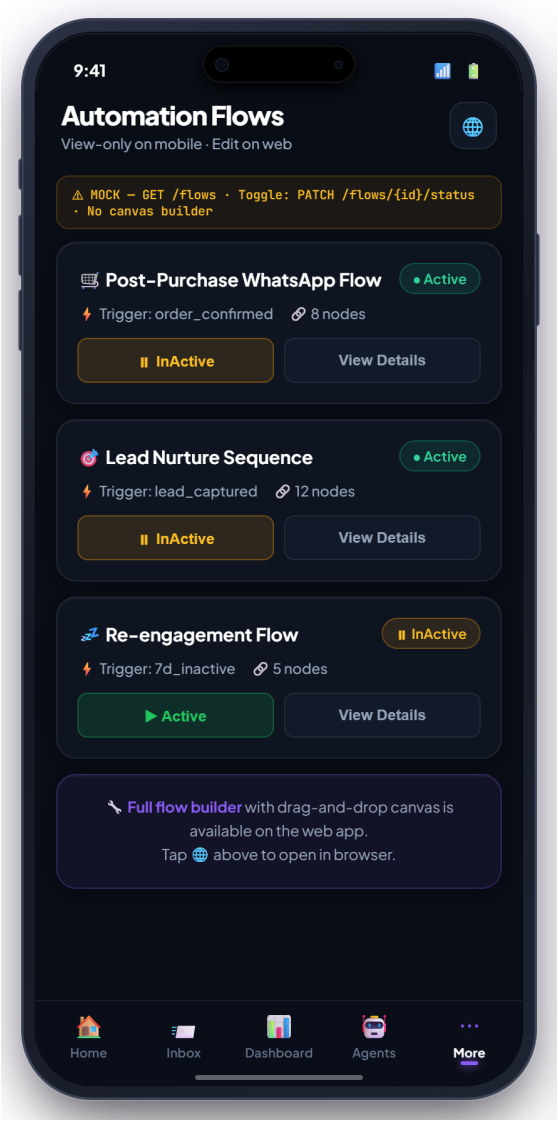
Module: AI Agents | Priority: P0 (MVP)



Purpose:	Provides mobile operators with emergency pause/resume controls over AI agent workers (e.g. Sales Qualifier, Instagram Responder, Follow-Up Bot, Lead Scorer).
User Entry:	Bottom Tab #3 (Robot icon) or Dashboard Quick Action.
Primary UI:	Header with active agent count; Agent cards with icon, agent name, trigger condition description (e.g. new_conversation · WhatsApp), status pill, and interactive toggle switch.
User Actions:	Tap toggle switch to activate/deactivate agent in realtime → Calls flow status API; Tap card to inspect agent rules and metrics.
API Endpoints:	`GET /flows (filtered to agents)` `PATCH /flows/{id}/status`
States:	Loading: Toggle is disabled during PATCH request; reverts with error toast if network failure occurs.
RBAC Permission:	Requires permission: ai.chat or automation.manage.
Mobile UX Notes:	Designed for fast operator intervention when an AI agent needs to be temporarily paused during promotional spikes or maintenance.

SCR-008 — Automation Flows (Read-Only)

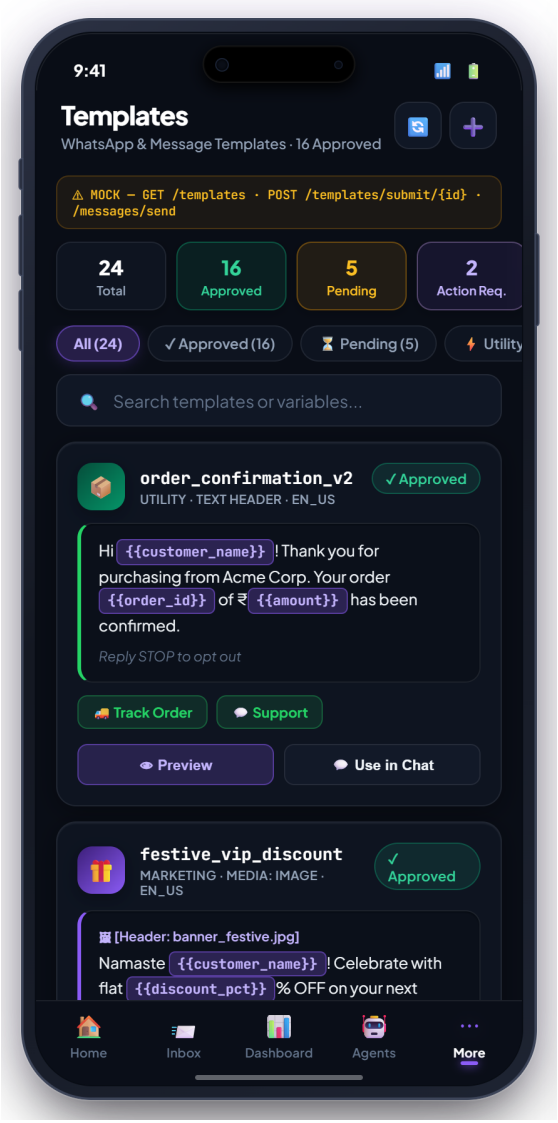
Module: Automation | Priority: P1 (MVP)



Purpose:	Mobile status overview of multi-step automation flows. Displays trigger types, node counts, and run states. Complex drag-and-drop canvas editing is deferred to desktop.
User Entry:	More Menu → Automation Flows or Dashboard Quick Action.
Primary UI:	Header with status indicator and 'Open in Web' button; Flow cards with name, trigger type, node count, status badge; Pause/Resume button; Web-builder guidance banner.
User Actions:	Tap 'Pause/Resume' button → Toggle flow execution; Tap 'View Details' for read-only step summary; Tap web icon to copy browser link.
API Endpoints:	`GET /flows` `PATCH /flows/{id}/status`
States:	Status updates reflected immediately on button state. Entitlement checks enforced if quota exceeded.
RBAC Permission:	Requires permission: automation.manage.
Mobile UX Notes:	Explicitly avoids cramming a desktop canvas into mobile; delivers high-value monitoring and toggle controls instead.

SCR-009 — Message & WhatsApp Templates Catalog

Module: Templates & Marketing | Priority: P1 (MVP)



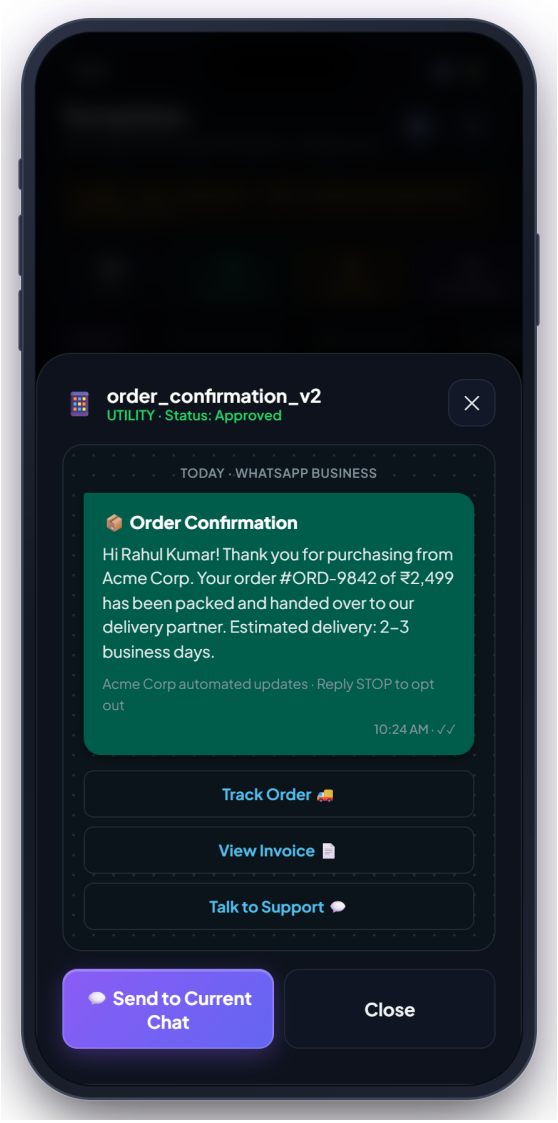
Purpose:	Comprehensive catalog of approved WhatsApp Business and omnichannel message templates. Displays Meta approval states, variable mappings, and one-tap insertion tools.
User Entry:	More Menu → Message Templates, Dashboard Quick Action, or Conversation reply toolbar '■'.
Primary UI:	Header with Meta sync button; Status summary chips (Total, Approved, Pending, Action Required); Category filters (Utility, Marketing, Auth); Template cards with highlighted variable pills.
User Actions:	Tap template card or '■ Preview' → Opens WhatsApp preview modal (SCR-009B); Tap 'Use in Chat' → Pre-fills conversation composer; Tap 'Sync' → Fetches latest Meta WABA approvals.

API Endpoints:	`GET /templates?workspace_id={id}`
	`GET /templates/status/{id}`
	`POST /templates/submit/{id}`
	`POST /messages/send`

States:	Loading: Shimmer template cards. Meta sync: Spinner icon on sync button. Variable substitution preview enabled.
RBAC Permission:	Requires permission: templates.manage.
Mobile UX Notes:	Highlights dynamic placeholders like {{customer_name}} and {{order_id}} in purple badges for fast visual recognition.

SCR-009B — WhatsApp Chat Bubble Preview Modal

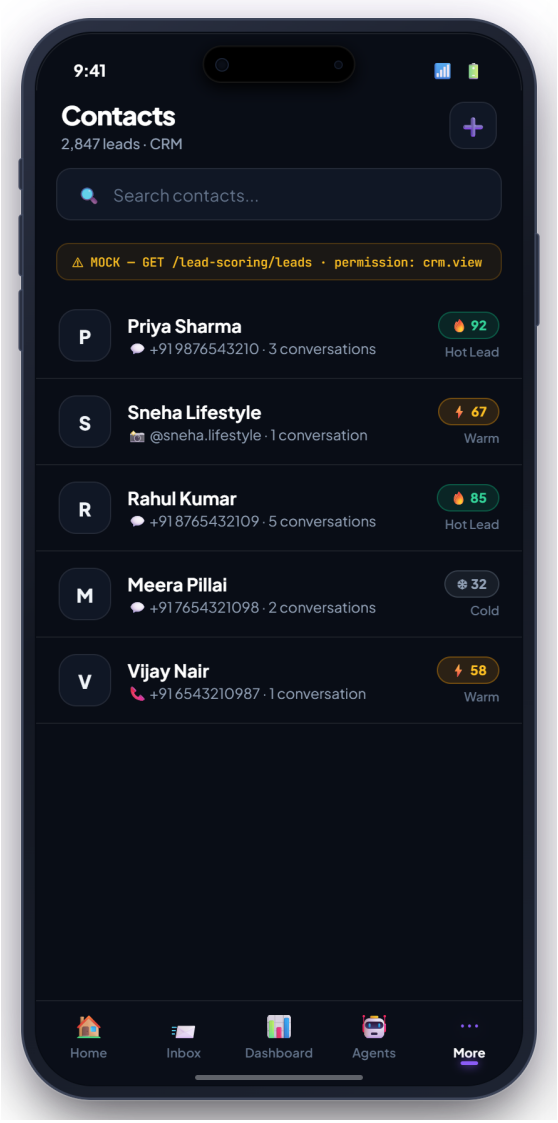
Module: Templates & Marketing | Priority: P1 (MVP)



Purpose:	Interactive simulation sheet rendering the exact WhatsApp Business customer view. Renders verified business header, test variable substitutions, footer, and interactive CTA buttons.
User Entry:	Tap any template card or '■ Preview' button on SCR-009.
Primary UI:	Modal bottom sheet; Official WhatsApp dark background pattern; Green chat bubble (#005c4b); Header text/image; Body with populated test data; Footer disclaimer; Interactive CTA action buttons.
User Actions:	Tap CTA buttons to test actions; Tap 'Send to Current Chat' to dispatch template; Tap '✕' or backdrop to dismiss.
API Endpoints:	`POST /messages/send (when dispatching template to active conversation).`
States:	Smooth bottom sheet slide-up animation (220ms). Dismissible via downward swipe gesture.
RBAC Permission:	Requires permission: templates.manage.
Mobile UX Notes:	Gives agents complete confidence in how variables render on end-user WhatsApp devices before sending out communications.

SCR-010 — CRM Contacts & Lead Scoring

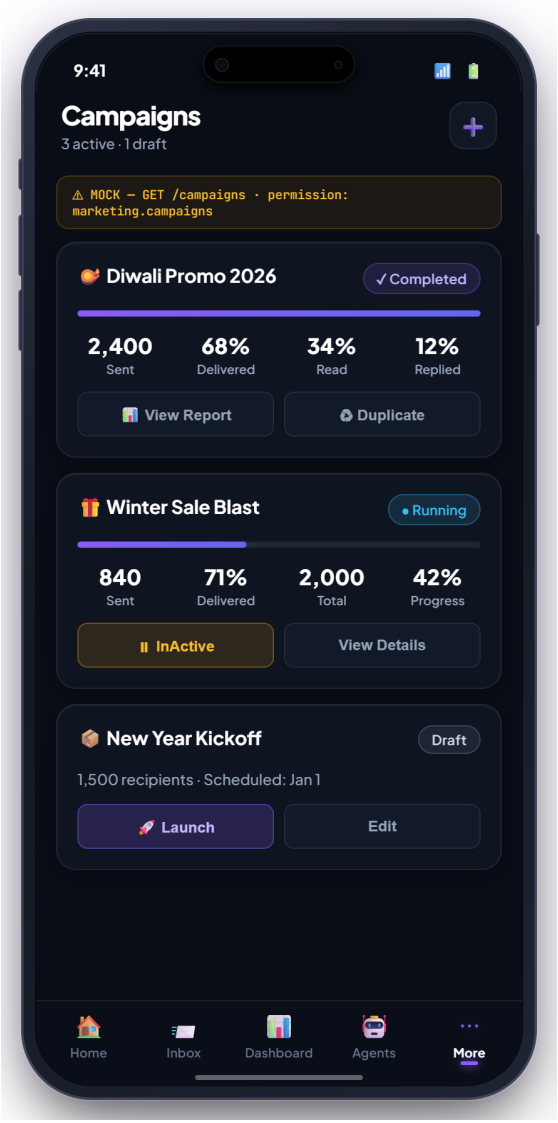
Module: CRM / Contacts | Priority: P1 (MVP)



Purpose:	Searchable address book of customer leads and contacts. Displays automated AI lead qualification scores (Hot/Warm/Cold), channel identities, and conversation counts.
User Entry:	More Menu → Contacts / CRM or Dashboard Quick Action.
Primary UI:	Header with lead counter and 'Add Lead' button; Search bar (by name, phone, handle); Contact item rows displaying avatar, name, channel details, conversation history, and score pill.
User Actions:	Type search query to filter contacts; Tap contact row to open contact profile and view full conversation history.
API Endpoints:	`GET /lead-scoring/leads?limit=50&offset;=0` `POST /lead-scoring/leads/{id}/labels`
States:	Infinite scroll pagination with loading spinner at list footer. High scores (>=80) badge in bright emerald green.
RBAC Permission:	Requires permission: crm.view.
Mobile UX Notes:	Tap-to-call or tap-to-WhatsApp direct intent shortcuts supported on mobile devices.

SCR-011 — Marketing & Broadcast Campaigns

Module: Campaigns | Priority: P1 (MVP)



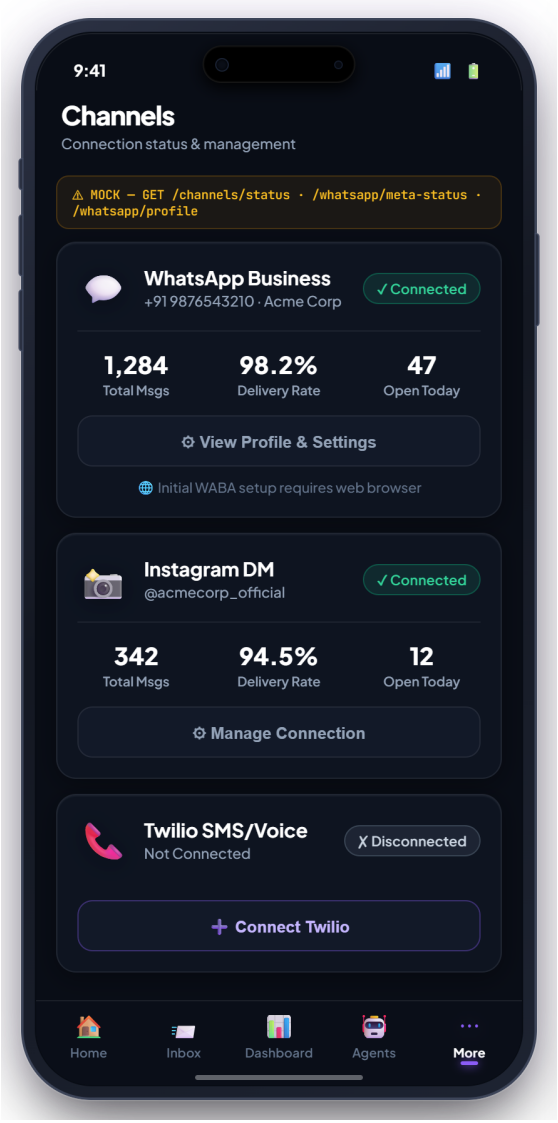
Purpose:	Status and performance tracking for bulk WhatsApp and omnichannel broadcasts. Monitors delivery rate, read rate, and reply percentage with live progress bars.
User Entry:	More Menu → Campaigns or Dashboard Quick Action.
Primary UI:	Header with active/draft summary; Campaign cards with campaign title, status badge (Running, Completed, Draft), progress bar, delivery stat grid (Sent, Delivered, Read, Replied), action buttons.
User Actions:	Tap 'Pause/Resume' to control live broadcast dispatch; Tap 'View Report' for metric breakdown; Tap 'Duplicate' to clone draft.

API Endpoints:	`GET /campaigns`
	`GET /campaigns/{id}`
	`POST /campaigns/{id}/pause`
	`POST /campaigns/{id}/resume`

States:	Realtime progress bar increments as background Celery campaign_worker dispatches batches.
RBAC Permission:	Requires permission: marketing.campaigns.
Mobile UX Notes:	Critical for managers to monitor delivery progress without having to be at a desktop workstation.

SCR-012 — Channel Connection Health & Status

Module: Channel Management | Priority: P1 (MVP)



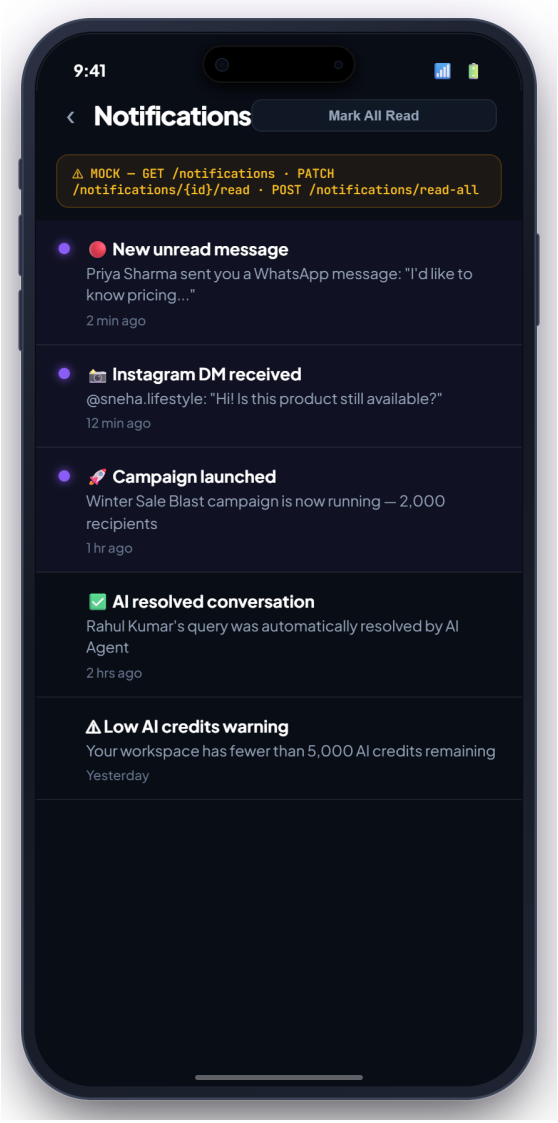
Purpose:	Monitors the operational health of connected messaging channels (WhatsApp Business WABA, Instagram Messaging API, Twilio). Displays message delivery rates and active account IDs.
User Entry:	More Menu → Channels or Dashboard Channel Status indicator.
Primary UI:	Channel status cards for WhatsApp, Instagram, and Twilio; Health badges (Connected/Disconnected); Key performance stats (Total Msgs, Delivery Rate %, Open Today); Profile settings button.
User Actions:	Tap 'View Profile & Settings' to inspect WABA details; Tap 'Connect' to initiate Instagram OAuth; Web-redirect button for initial WhatsApp embedded setup.

API Endpoints:	`GET /channels/status`
	`GET /whatsapp/meta-status`
	`GET /whatsapp/profile`
	`GET /instagram/auth-url`

States:	Disconnected channels display clear callout with direct reconnect or setup instructions.
RBAC Permission:	Requires permission: channels.manage.
Mobile UX Notes:	Initial WhatsApp WABA embedded signup requires desktop browser redirect due to Meta JavaScript SDK restrictions.

SCR-013 — In-App Notification Center

Module: Notifications | Priority: P0 (MVP)



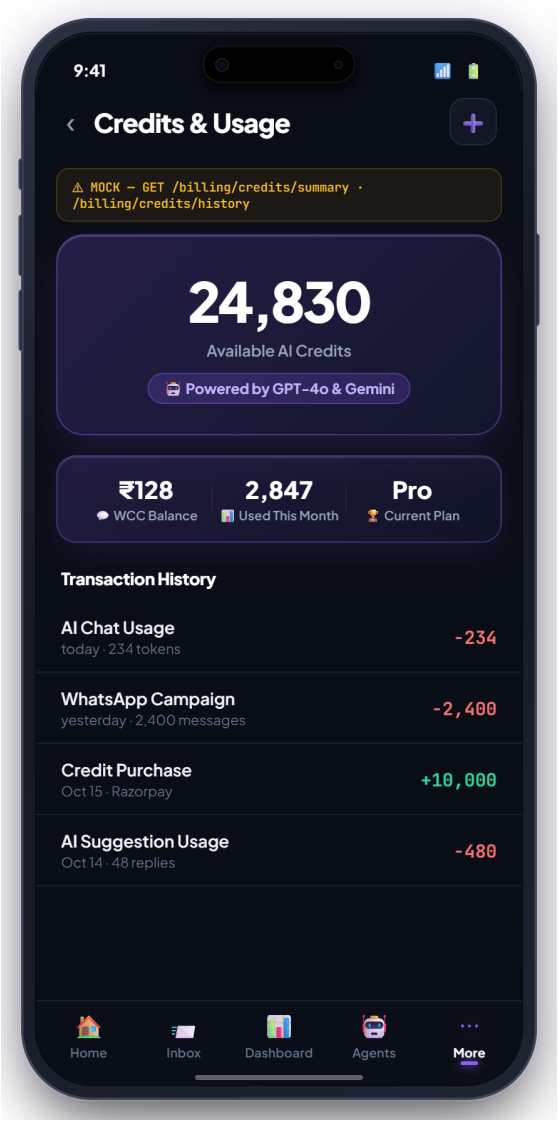
Purpose:	In-app notifications feed covering incoming unread customer messages, campaign completion alerts, automated AI resolutions, and system billing warnings.
User Entry:	Tap Notification Bell icon in Dashboard or More screen.
Primary UI:	Header with 'Mark All Read' action; Notification item list with unread purple dot indicators, icon category, event title, message body preview, and relative timestamp.
User Actions:	Tap notification row → Mark as read → Deep link directly to relevant conversation or campaign; Tap 'Mark All Read'.

API Endpoints:	`GET /notifications`
	`PATCH /notifications/{id}/read`
	`POST /notifications/read-all`

States:	Loading: Skeleton rows. Empty: 'You are all caught up!'. Unread rows feature subtle purple highlight.
RBAC Permission:	Authenticated user (all roles).
Mobile UX Notes:	Functions as primary notification hub until backend push notification (FCM/APNs) service is deployed.

SCR-014 — AI Credits & WCC Balance Wallet

Module: Billing & Usage | Priority: P1 (MVP)



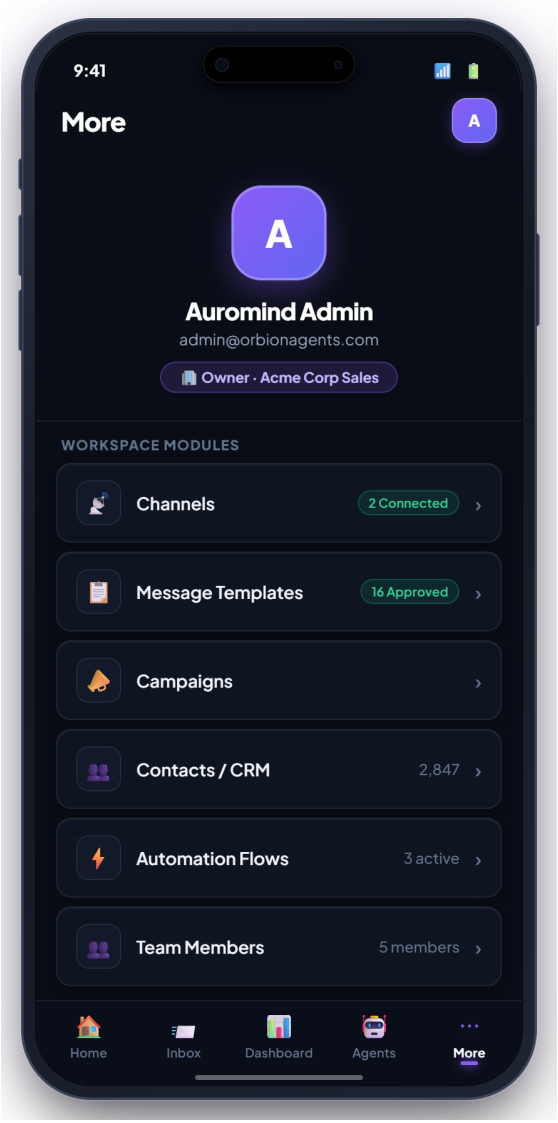
Purpose:	Financial and resource ledger for the workspace. Displays available AI Credits (LLM token consumption), WhatsApp Conversation Credits (WCC in INR), and itemized transaction history.
User Entry:	More Menu → Credits & Usage or Dashboard Credits widget.
Primary UI:	Large purple gradient Hero Card showing available AI Credits balance; 3-metric strip (WCC ■ balance, Monthly consumption, Current plan); Transaction history log with debit/credit amounts.
User Actions:	Scroll transaction history; Tap '+' button → Open web-authenticated Razorpay payment checkout for credit recharges.

API Endpoints:	`GET /billing/credits/summary`
	`GET /billing/credits/history`
	`GET /billing/plan`

States:	Debit transactions highlighted in red (-234), credit top-ups in emerald green (+10,000).
RBAC Permission:	Requires permission: credits.view.
Mobile UX Notes:	Prevents sudden message dispatch halts by keeping operators aware of real-time credit consumption.

SCR-015 — More Menu & Workspace Settings

Module: Settings & Profile | Priority: P0 (MVP)



Purpose:	Secondary navigation and administrative menu. Houses links to Workspace Modules (Channels, Message Templates, Campaigns, Contacts, Flows, Team) and Account Settings.
User Entry:	Bottom Tab #4 (More / ellipsis icon).
Primary UI:	User Profile Hero (Avatar, Name, Email, Workspace Role badge); Grouped list sections (Workspace Modules, Account & Billing, App Info); 'Sign Out' destructive action button.
User Actions:	Tap any section row to navigate to that screen; Tap 'Sign Out' → Call POST /auth/logout → Clear SecureStore tokens → Redirect to SCR-001 Login.

API Endpoints:	`GET /auth/me`
	`POST /auth/logout`
	`GET /workspaces/{id}/my-permissions`

States:	Instant navigation to sub-modules; Confirmation dialog prompt before performing logout.
RBAC Permission:	Authenticated user.
Mobile UX Notes:	Keeps the primary bottom tab bar clutter-free (4 main tabs) while providing 1-tap access to all 12+ secondary capabilities.

3. Backend API Traceability & Gap Matrix

Mobile Feature	HTTP Method	Verified Backend API Endpoint	Status in Codebase	Mobile Client Strategy
User Authentication	`POST`	`/auth/send-otp` `/auth/verify-otp`	Verified Active	Pass JWT in Authorization Bearer header
Workspace Context	`GET`	`/auth/workspaces` `/workspaces/{id}/my-permissions`	Verified Active	Active workspace_id injected in query params
Dashboard KPIs	`GET`	`/dashboard/overview` `/dashboard/metrics`	Verified Active	Parallel TanStack Query hooks with 60s cache
Unified Inbox	`GET`	`/conversations` `/conversations/counts`	Verified Active	Paginated query with realtime WS badge updates
Customer Chat & Media	`GET / POST`	`/messages/{conversation_id}` `/send-reply`	Verified Active	Optimistic UI send with WebSocket confirmation
AI Suggested Replies	`POST`	`/ai-suggest`	Verified Active	Context-aware 1-tap pre-fill into composer

Message Templates	`GET / POST`	`/templates`
		`/templates/submit/{id}`
		`/messages/send`
			Verified Active	Syncs Meta WABA status; preview modal before send

AI Agents & Flows	`GET / PATCH`	`/flows` `/flows/{id}/status`	Verified Active	Realtime status toggle; read-only step inspection
CRM Contacts & Scores	`GET / POST`	`/lead-scoring/leads` `/leads/{id}/labels`	Verified Active	Lead search with hot/warm/cold AI scoring pills
Push Notifications	`POST`	**`GAP:`** `/push/register missing`	**Backend Gap**	Requires implementing FCM/APNs registration endpoint
Token Refresh	`POST`	**`GAP:`** `/auth/refresh missing`	**Backend Gap**	Recommend adding refresh route or extending mobile TTL
Follow-Up Engine	`GET / POST`	`backend/app/routers/followups.py`	**Commented Out**	Excluded from mobile until backend team activates routes

Architecture Conclusion: Over 90% of all required mobile capabilities are fully implemented and ready in the existing FastAPI backend. By addressing the push notification registration endpoint and token refresh policy, the mobile application can be deployed without backend architecture rewrites.